Answers:

1. a) $y = -x + 3$

b) $y = -\frac{2}{3}x - 2$

c) $y = \frac{1}{4}x + 3$

d) $y = -\frac{3}{2}x + \frac{5}{2}$

2. a) slope -1; y-intercept 3; the graph is a line crossing the y-axis at 3 and the x-axis at 3.

b) slope $-\frac{2}{3}$; y-intercept -2; the graph is a line crossing the y-axis at -2 and the x-axis at -3.

c) slope $\frac{1}{4}$; y-intercept 3; the graph is a line crssing the y-axis at 3 and passing through (4, 4).

d) slope $-\frac{3}{2}$; y-intercept $\frac{5}{2}$; the graph is a line crossing

the y-axis at $2\frac{1}{2}$ and passing through $(3, -2)$.

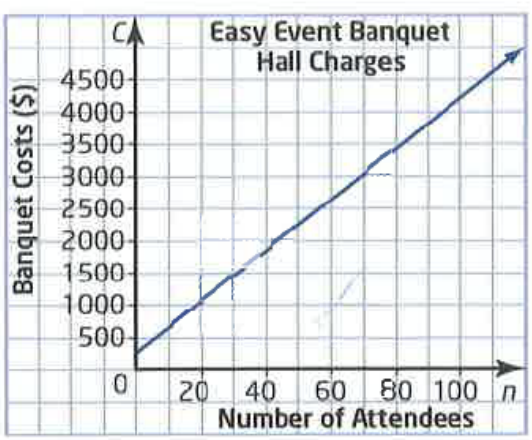
3. a) slope $-\frac{1}{3}$; y-intercept 1

b) slope $\frac{2}{5}$; y-intercept $\frac{8}{5}$

4. a) $C = 40n + 250$

b) fixed cost \$250; variable cost \$40 per person

c)



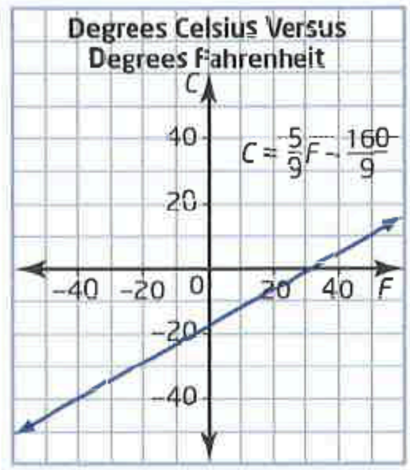
d) \$4250

5. \$15, \$20

6.

a) $C = \frac{5}{9}F - \frac{160}{9}$

b)



c) The slope is $\frac{5}{9}$ and the C-intercept is $-\frac{160}{9}$. The slope is a multiplication coefficient and the C-intercept is a constant. To change a Fahrenheit temperature to a Celsius temperature, multiply the Fahrenheit temperature by the slope and add the C-intercept.

7. a) $C = 25n + 1250$; $C = 30n + 995$
   b) Knights: fixed = 1250, variable=25
      Legions: fixed=995, variable=30
   c) Knights=\$2375 ; Legions=\$2345

8. a) 2x+y-7=0; A=2, B=1, C=-7
   b) 3x-4y-8=0; A=3, B=-4, C=-8